



MD SAAD KHAN

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Kolkata, West Bengal, India

[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [LeetCode](#)

PROFILE

Electronics student at IIT Madras with strong foundation in signal processing, embedded systems, and Edge-AI. Experienced in designing filtering circuits, deploying CNN models on edge devices, and working with FPGA-based systems. Interested in wireless communication systems and signal analysis research.

Highlights

- FPGA, Digital IC, and electronic system design
- Embedded systems, IoT, and sensor interfacing
- Efficient and structured software development

TECHNICAL SKILL

- **Programming:** Python, Verilog, Assembly, C, JavaScript
- **Hardware:** FPGA Design, Digital Electronics, Circuit Design
- **Tools:** Xilinx Vivado, ONNX, Linux (Ubuntu), Git, GitHub, Pixelpulse2
- **Concepts:** Signal Processing, Embedded Systems, Communication Basics

LANGUAGE KNOWN

- **English**
Full professional proficiency
- **Hindi**
Bilingual proficiency
- **Bengali**
Limited working proficiency
- **Urdu**
Limited working proficiency

EDUCATION

Indian Institute of Technology Madras

Chennai, Tamil Nadu – 600036

Diploma in Electronic Systems (Ongoing | Expected: 2026)

Bachelor of Science, BS in Electronic Systems (Integrated program)

CGPA: 7.5 / 10 (as of 2026)

- Strong foundation in digital electronics, electrical and electronic circuit, network theory, C, Linux and problem-solving.

St. Ann's English Medium School

Howrah, West Bengal - 711101

WBCHSE, PCMB (2021 – 2023) [view](#)

Grade: AA

- Completed high school with a strong academic background, fostering analytical thinking and problem-solving skills.

WORK EXPERIENCE

Summer Research Internship

(Jun'26 – Jul'26)

Department of Computer Science & Engineering, RV University Bangalore

- Conducted cybersecurity literature reviews and analyzed recent research on emerging threats and defense mechanisms.

Software Development Intern

(Apr'26 – May'26)

Bluestock™ Fintech

- Assisted in the development, testing, and maintenance of fintech applications, contributing to feature implementation, debugging, and software quality improvement.

Research Intern

(Jan'26 – Mar'26)

CyStar Lab - Centre for Cybersecurity, Trust and Reliability (CS14)

Department of Computer Science & Engineering, IIT Madras

- Conducted research on Hardware Security Modules (HSMs), analyzing cryptographic key management, security architectures, and deployment practices through literature review and technical evaluation.

Data Analytics Intern

(Jun'25 – Jan'26)

InternSpark Pvt Limited

- Performed data analysis, reporting, and visualization tasks to extract actionable insights and support data-driven decision-making processes.

KEY ACADEMIC PROJECTS & ASSIGNMENTS

- **Multi-Digit Counter and Seven-Segment Display System | Course Assignment**

Course: EE1902 - Electronics Lab | Instructor: Prof. Janakiraman Viraraghavan, EE Dept., IIT Madras

- Designed and implemented a 4-bit synchronous counter-based decimal display system using 74LS163, 74LS85, 74LS83, and 74LS47 ICs; developed custom logic for two-digit seven-segment display visualization without dedicated driver ICs.
- Integrated BJT-based LED driving circuits, analyzed IC output current limitations, and validated circuit performance through hardware testing, voltage/current measurements, and debugging using ADALM1000 (M1K).
- **Design & Verification of Dual-Clock Asynchronous FIFO | Self-Project**
 - Designed a 64×8 asynchronous FIFO in Verilog for data transfer across independent clock domains.
 - Implemented Gray-coded pointers and 2-flop synchronizers for reliable clock domain crossing.
 - Verified Empty conditions using testbench simulation (50 MHz write, 10 MHz read) in GTKWave.
- **Mail ID & Password Generator (v1.0.2) | CS50 - Harvard University Final Project - [Repo](#) | [Live Preview](#)**
 - Built an interactive HTML-based tool to generate unique email IDs and secure passwords with copy-to-clipboard functionality and a clean, efficient UI.
 - Upgraded the generator to meet organizational formatting standards for Data Entry roles, integrating real-world usability with academic innovation.
- **India Electronics and Semiconductor Association (IESA) DeepTech Hackathon 2026 - [Repo](#)**
 - Developed a lightweight Edge-AI solution for semiconductor wafer defect classification during the IESA DeepTech Hackathon 2026, trained a custom CNN to classify 8 defect categories with ~90% validation accuracy.
 - Optimized the model for edge deployment by exporting to ONNX and applying quantization techniques, targeting NXP i.MX RT series microcontrollers.
- **IITM BS Course Assistant – RAG Pipeline | Self-Project - [view](#)**
 - Developed an interactive RAG assistant to query the IITM BS Student Handbook/Grading using LangChain and the Mistral-7B LLM.
 - Engineered a custom embedding pipeline using the HuggingFace API and a FAISS vector database to efficiently chunk, index, and retrieve document context.

HONORS & AWARDS

- **Global Quantum Mechanics Challenge, 2026- [view](#)**
 - 3rd Prize (Senior) with a \$125 cash prize; achieved Gold Honour — Top 1% overall globally.
 - Successfully cleared the Qualification, Semi-Final, and Final Rounds of the competition.
- **Tata Capital Scholarship (2025–26) - [view](#)**
 - Awarded the Tata Capital Pankh Scholarship (2025-26) by the Buddy4Study India Foundation.
 - Secured competitive financial aid in recognition of academic excellence to support undergraduate studies.
- **International Physics Competition (IPhYC) 2025 - [view](#)**
 - Gold Honour; **Top 1%** globally (~5,600 participants)
 - Final Round: **Top 7%** (~960 participants), Score: 38
 - Qualified through all stages; awarded 100% fee waiver

CERTIFICATIONS

- **Introduction to Hugging Face | 365 DataScience - 2026 [view](#)**
- **FPGA to ASIC: A Complete VLSI Design Flow | NIT Rourkela (IEEE) – 2026 [view](#)**
- **CS50-Introduction to Computer Science | Harvard University – 2025 [view](#)**
- **Remote Sensing Data Analytics | ISRO – 2025 [view](#)**
- **Master in 3D Animation course | Professor of How – 2024 [view](#)**

EXTRACURRICULARS & HOBBIES

- **Game Modding:** Actively mod *GTA V*, *GTA San Andreas* and *GTA Vice City*, working with game assets, scripts, and custom modifications to enhance gameplay and visuals.
- **Technical Exploration:** Exploring embedded systems, electronics projects, and new tools in AI and software development.
- **Hobbies:** Watching movies, listening to music, cricket, and playing table tennis & badminton.